Marshall Elford

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TECHNICAL SKILLS

Computer

- Java
- C/C++
- HTML/XML
- Verilog
- Assembly (ARM/x86_64)
- MATLAB
- Git
- Microsoft Office

Communication

- Stage Management
- Team Leading
- Public Speaking
- Debate Team
- Technical Writing

Electrical

- Oscilloscope
- Multimeter
- Function Generator
- Breadboard
- Hand Tools

TECHNICAL WORK EXPERIENCE

PJS Systems **April, 2016 – August, 2016**
Audio/Visual Systems Installer

- Installed audio and visual systems for the University of British Columbia's Aquatic Centre, Langara College's Science & Technology building, and Translink Head Office.
- These systems included digital displays, full room control units, and public announcement systems.
- Communicated effectively with other team members, as well as worked independently, to problem solve and implement technical solutions.
- Trained and proficient in the use of many technical tools including wire trimmers, plyers, screwdrivers, X-ACTO knives, and electrical tape.

TECHNICAL PROJECTS

Multi-Functional Robot **March, 2017**

- Implemented a robot using C with the ability to autonomously drive, turn, and act as a functional printer.
- Designed and implemented the ability to drive straight using hall-effect sensors, detecting magnets on each wheel to verify equal contribution from each DC motor. Mounted the range finder sensor on the servo motor to continuously detect objects and determine the safest direction to drive.
- Added the ability to print black and white pictures by converting them to binary, using the binary string to command a servo motor with a pen attached when to raise and lower, and commanding the robot to step through each bit.
- Worked with a team to complete the design with the use of Git to allow collaboration between partners.

Android App with Hardware Communication **March, 2017**

- Designed an app using Android Studio (Java/XML) that allows users to input items with IDs, each connected to a weight scale, that tracks how much of the item is left and notifies the user when the item is about to run out.
- Connected the scales to an Arduino, which communicated with a Raspberry Pi 3 and used Wi-Fi to send the data to the app and a connected web server.

Virtual World with AI **November, 2016**

- Using Java, created a virtual world involving class and interface hierarchy to simulate the interaction of AI objects.
- Designed intelligent AI classes with defined characteristics and actions.
- Worked with a partner using Git to collaborate and track each other's progress.
- Displayed the randomly generated world using Microsoft Virtual Studio

RISC Machine **October, 2016**

- Using Verilog, coded a simple RISC machine, able to take in instructions, perform functions, and output results.

- Operated the machine and displayed results on a DE1-SOC.

HTML Validator **October, 2016**

- Designed a program in Java that read HTML code and analyzed the syntax, using an algorithm to identify and explain incorrect tags throughout.
- Implemented a stack used to store tags and allow the algorithm to compare tags.

Social Network Analyzer **October, 2016**

- Designed a program in Java that analyzes a Twitter dataset with over 10 000 user entries.
- Created graph implementations of an adjacency list and an adjacency matrix, used for creating breadth and depth first search algorithms that identify common influencers between users and the number of retweets required for a user to appear on another user's feed.

Autonomous Claw Machine **February, 2016 – March, 2016**

- Created a functioning autonomous claw using an Arduino Uno, able to successfully pick up and drop a wide assortment of objects
- Worked collaboratively in a team of 6 to mutually develop the project from start to finish.
- Handled Engineering tools to cut sheet metal and develop a prototype from a virtual model.
- Performed the fastest out of every competing team, picking up and placing four objects in under 18 seconds.

VOLUNTEER WORK EXPERIENCE

Theatre Department **November, 2011 – April, 2015**

Backstage Manager

- Led a group of 30+ actors and a backstage crew, every year for four years.
- Communicated effectively with individuals, instructing specific movement and action to be portrayed.
- Organized and presented a 3+ hour performance, directing from behind the scenes and ensuring all complications be settled in a timely manner.

EDUCATION

University of British Columbia **September, 2015 – Present**

Bachelor of Applied Science - Computer Engineering

AWARDS

American Mathematics Competition 12 School Champion **2015**

Province of B.C. Grade 12 Provincial Scholarship Award for Excellence **2015**

PROFESSIONAL AFFILIATIONS

IEEE – Professional Association for the Advancement of Technology

ACTIVITIES AND INTERESTS

- Soccer, Baseball, Swimming
- Programming, Design, Testing
- Visual Arts (Film, Editing)